

Infinitesimal Calculus 2, Moed B

2022, Reichman University Solutions

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Instructions:

1. Duration: 3 hours.
2. Only a **basic** calculator and the attached formula sheets are permitted in the exam.

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The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. It is forbidden to rely on any material that was not learned in the course, unless it is proven. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. (25 points) Let $x_0, x_1 \in \mathbb{R}$ and let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers.

- (a) (3 points) Define the term: the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely at x_1 . You must write your definition in the ϵ - N language.

Solution: We say that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely at x_1 if there exists an $s \in \mathbb{R}$ such that for every $\epsilon > 0$ there exists a $k \in \mathbb{N}$ such that for every $N \geq k$ we have

$$\left| \sum_{n=0}^N |a_n(x_1 - x_0)^n| - s \right| < \epsilon$$

- (b) (15 points) Suppose that there exists an $0 \leq L \leq \infty$ such that $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Let $0 \leq R \leq \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$. Prove that if $0 < L < \infty$ then $R = \frac{1}{L}$.

- (c) (7 points) Compute the value of the following definite integral:

$$\int_0^1 x \ln(1 + x^2) dx$$

Solution: Substituting $t = \ln(1 + x^2)$ then $dt = \frac{2x}{1+x^2} dx$, and

$$\int_0^1 x \ln(1 + x^2) dx = \int_0^{\ln(2)} \frac{xt \cdot (1 + x^2)}{2x} dt = \frac{1}{2} \int_0^{\ln(2)} te^t dt = \frac{1}{2} [te^t - e^t]_0^{\ln(2)} = \ln(2) - \frac{1}{2}.$$

2. (25 points)

- (a) (10 points) Let $a, b \in \mathbb{R}$ such that $a < b$. Let f be a function that is defined on $[a, b]$. Suppose that f is continuous on $[a, b]$ and let G be an anti-derivative of f in $[a, b]$. Prove that

$$\int_a^b f(x) dx = G(b) - G(a)$$

- (b) (15 points) Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a monotonically decreasing function, such that $\lim_{x \rightarrow \infty} f(x) = 0$.

- i. (6 points) Prove that there exists a $0 \leq k \in \mathbb{Z}$ for which the function $\sin(f(x))$ is decreasing on $[k, \infty)$.

Solution: As f is decreasing then $\inf \{f(x) : x \geq 0\} = \lim_{x \rightarrow \infty} f(x) = 0$, and therefore $f(x) \geq 0$ for every $x \geq 0$. Also, as $\lim_{x \rightarrow \infty} f(x) = 0$, then there exists an $M > 0$ such that for every $x > M$ we have $f(x) \leq 1$. Thus, for every $x > M$, $0 \leq f(x) \leq 1$. Let $x_1, x_2 > M$ such that $x_1 < x_2$. Then $0 \leq f(x_2) \leq f(x_1) \leq 1$ and therefore $\sin(f(x_2)) \leq \sin(f(x_1))$, which implies that $\sin(f(x))$ is decreasing on (M, ∞) . Choosing any $k \in \mathbb{N}$ such that $k > M$ (by the Archimedean property, such a k exists), then we obtain that $\sin(f(x))$ is decreasing on $[k, \infty)$.

- ii. (9 points) Suppose that the integral $\int_0^{\infty} \sin(f(x)) dx$ is convergent. Prove that the series $\sum_{n=0}^{\infty} (e^{f(n)} - 1)$ is convergent.

Solution: First, as $\int_0^{\infty} \sin(f(x)) dx$ is convergent then $\int_k^{\infty} \sin(f(x)) dx$ is convergent. As $\sin(f(x))$ is decreasing on $[k, \infty)$ and as $\lim_{x \rightarrow \infty} \sin(f(x)) = \sin(0) = 0$, then by the integral test, the series $\sum_{n=k}^{\infty} \sin(f(n))$ is convergent. Applying Heine's Lemma and L'hospital's rule, we get

$$\lim_{n \rightarrow \infty} \frac{\sin(f(n))}{e^{f(n)} - 1} = \lim_{t \rightarrow 0} \frac{\sin(t)}{e^t - 1} = \lim_{t \rightarrow 0} \frac{\cos(t)}{e^t} = 1$$

Thus, by the limit comparison test, the series $\sum_{n=0}^{\infty} (e^{f(n)} - 1)$ is convergent.

3. (25 points)

- (a) (13 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive real numbers. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the series $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ is convergent.

Solution: Suppose first that the series $\sum_{n=1}^{\infty} a_n$ is convergent. Then by the comparison principle for nonnegative series we have

$$\sum_{n=1}^{\infty} \frac{a_n}{1+a_n} \leq \sum_{n=1}^{\infty} a_n < \infty$$

implying that $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n} < \infty$.

Suppose now that $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n} < \infty$. Then by the necessary criterion, we have $\lim_{n \rightarrow \infty} \left[1 - \frac{1}{1+a_n}\right] = \lim_{n \rightarrow \infty} \frac{a_n}{1+a_n} = 0$, and thus $\lim_{n \rightarrow \infty} \frac{1}{1+a_n} = 1$. Note that $\lim_{n \rightarrow \infty} \left[\frac{a_n}{1+a_n}\right] = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{1+a_n}} = 1$ and therefore, by the limit comparison test, the series $\sum_{n=1}^{\infty} a_n$ is convergent.

- (b) (12 points) Let $(a_n)_{n=0}^{\infty}, (b_n)_{n=0}^{\infty}$ be two sequence of real numbers. Suppose that $|b_n| \leq 2$ for every $0 \leq n \in \mathbb{Z}$. Let $0 \leq R_1 \leq \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n x^n$ and let $0 \leq R_2 \leq \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n b_n x^n$.

- i. (6 points) Prove that $R_2 \geq R_1$.

Solution: Let $|x| < R_1$. Then the power series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely at x . By the comparison principle, we have $\sum_{n=0}^{\infty} |a_n b_n| |x|^n \leq \sum_{n=0}^{\infty} 2 |a_n| |x|^n = 2 \sum_{n=0}^{\infty} |a_n| |x|^n < \infty$. Thus, the power series $\sum_{n=0}^{\infty} a_n b_n x^n$ converges absolutely for every $|x| < R_1$, which implies that $R_2 \geq R_1$.

- ii. (6 points) Give an example where $R_2 > R_1$.

Solution: We choose $a_n = 1$ and $b_n = \frac{1}{2^n}$ for every $0 \leq n \in \mathbb{Z}$. Then the power series $\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} x^n$ converges if and only if $|x| < 1$ (geometric), which implies that $R_1 = 1$. The power series $\sum_{n=0}^{\infty} a_n b_n x^n = \sum_{n=0}^{\infty} \frac{x^n}{2^n} = \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$ converges if and only if $\left|\frac{x}{2}\right| < 1$, or $|x| < 2$, which implies that $R_2 = 2$. Thus, $R_2 > R_1$.

4. (25 points)

- (a) (12 points) Prove that the series $\sum_{n=1}^{\infty} [n(1-p)^{n-1} \cdot p]$ converges for every $0 < p < 2$, and compute its value as a function of p .

Solution: Setting $x = 1 - p$, then $0 < p < 2$ implies that $-1 < x < 1$ and that $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. Thus, using term-by-term differentiation, we obtain

$$\sum_{n=1}^{\infty} [n(1-p)^{n-1}] = \sum_{n=0}^{\infty} n x^{n-1} = \left[\frac{1}{1-x} \right]' = \frac{1}{(1-x)^2} = \frac{1}{p^2}.$$

We conclude that $\sum_{n=1}^{\infty} [n(1-p)^{n-1} \cdot p] = p \sum_{n=1}^{\infty} [n(1-p)^{n-1}] = \frac{p}{p^2} = \frac{1}{p}$.

- (b) (13 points)

- i. (4 points) Prove that for every $q > 0$ there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $\ln(n) \leq n^q$.

Solution: Let $q > 0$. We have by Heine's lemma and L'Hopital's rule,

$$\lim_{n \rightarrow \infty} \frac{\ln(n)}{n^q} = \lim_{x \rightarrow \infty} \frac{\ln(x)}{x^q} = \lim_{x \rightarrow \infty} \frac{1}{q x^{q-1} \cdot x} = \lim_{x \rightarrow \infty} \frac{1}{q x^q} = 0.$$

Thus, there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $\frac{\ln(n)}{n^q} < 1$, which implies that $\ln(n) \leq n^q$.

- ii. (9 points) Find all values of $r \in \mathbb{R}$ for which the following series converges: $\sum_{n=3}^{\infty} \frac{\ln(n)}{n^r}$.

Solution: Note that for $r \leq 1$ we have

$$\sum_{n=3}^{\infty} \frac{\ln(n)}{n^r} \geq \sum_{n=3}^{\infty} \frac{\ln(n)}{n} \geq \sum_{n=3}^{\infty} \frac{1}{n} = \infty$$

as it is a tail of the Harmonic series. Therefore, by the comparison principle we have $\sum_{n=3}^{\infty} \frac{\ln(n)}{n^r} = \infty$. Suppose now that $r > 1$ and let $q = \frac{r-1}{2}$. Then $r - q = \frac{1+r}{2} > 1$. By the previous item, there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $\ln(n) \leq n^q$. Thus

$$\sum_{n=k}^{\infty} \frac{\ln(n)}{n^r} \leq \sum_{n=k}^{\infty} \frac{n^q}{n^r} = \sum_{n=k}^{\infty} \frac{1}{n^{r-q}} < \infty$$

as the $r - q$ Harmonic series is convergent.

5. (25 points)

(a) (15 points) Prove that for every $-1 \leq x \leq 1$ we have

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

Solution: Setting $t = -x^2$, then for every $|t| < 1$ we have

$$\frac{1}{1+x^2} = \frac{1}{1-t} = \sum_{n=0}^{\infty} t^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

Note that $|t| = x^2 < 1$ if and only if $-1 < x < 1$, and therefore the above equality holds at least for every $-1 < x < 1$. Integrating both sides, then there exists a $C \in \mathbb{R}$ such that

$$\arctan(x) + C = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

Plugging $x = 0$ yields $C = 0$, and therefore

$$\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

for every $-1 < x < 1$. Note that the series $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ is convergent for $x = 1$ (as it is a negative of a Leibniz series) and is convergent for $x = -1$ (as it is a Leibniz series). Thus,

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2n+1} = \lim_{x \rightarrow 1^-} \arctan(x) = \arctan(1)$$

and

$$\sum_{n=0}^{\infty} (-1)^n \frac{-1}{2n+1} = \lim_{x \rightarrow -1^+} \arctan(x) = \arctan(-1)$$

and we obtain that

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

for every $-1 \leq x \leq 1$.

(b) (3 points) Let $D \subseteq \mathbb{R}^2$, let $(x_0, y_0) \in D$ and let $F : D \rightarrow \mathbb{R}$ be a function that is defined in a neighbourhood of (x_0, y_0) . Define the term: $\nabla F(x_0, y_0)$.

Solution: We define

$$\nabla F(x_0, y_0) = \left(\lim_{x \rightarrow x_0} \frac{F(x, y_0) - F(x_0, y_0)}{x - x_0}, \lim_{y \rightarrow y_0} \frac{F(x_0, y) - F(x_0, y_0)}{y - y_0} \right)$$

(c) (7 points) Prove or disprove the following statement: the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ that is defined by

$$F(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

has a directional derivative at $(0, 0)$ in every direction.

Solution: False. Consider the direction $\vec{u} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$. Then

$$D_{\vec{u}}(F)(0, 0) = \lim_{h \rightarrow 0} \frac{F\left(\frac{h}{\sqrt{2}}, \frac{h}{\sqrt{2}}\right) - F(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^2}{2h^3} = \lim_{h \rightarrow 0} \frac{1}{2h} = DNE$$